

AI Concepts Report

This report answers the five questions in simple original wording and includes a generated AI-style illustration.

Q1. What is Artificial Intelligence?

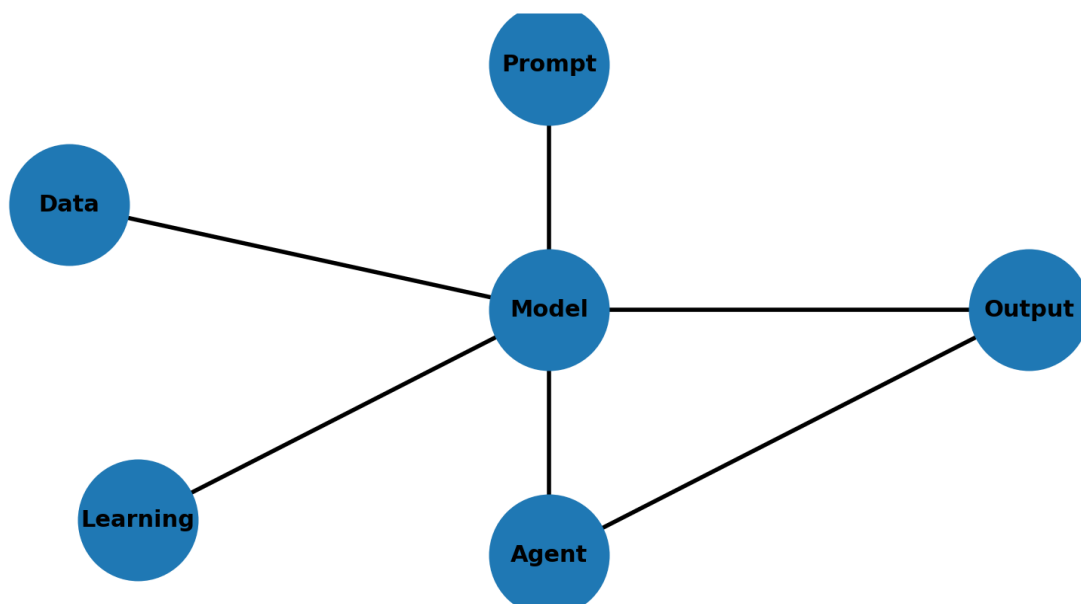
Artificial Intelligence, or AI, is the field of building computer systems that can do tasks normally associated with human thinking. These tasks include understanding language, recognizing patterns, learning from experience, solving problems, and making decisions. Instead of only following fixed instructions, an AI system can use data to improve its behavior and produce smarter results over time.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

AI is the broad umbrella term for machines that show intelligent behavior. Machine Learning is one part of AI where the system learns patterns from data rather than being explicitly programmed for every rule. Generative AI is a type of AI that creates new content such as text, images, music, or code. Agentic AI goes further by not only generating content, but also planning steps, using tools, and acting toward a goal with less human supervision. So, the relationship is like this: AI is the whole field, ML is a learning method inside AI, generative AI is a content-creation style inside AI, and agentic AI focuses on taking actions to complete tasks.

AI-Generated Illustration

AI Workflow Illustration



The illustration above is a generated visual that represents how data, prompts, learning, models, outputs, and agents connect in an AI system.

Q3. Explain any three open-source AI models available on Hugging Face

1) GPT-2 - GPT-2 is a well-known open language model from OpenAI that is available in open-source form on Hugging Face. It is useful for text generation, completion, and basic language experiments. Many students and developers use it because it is lightweight compared with newer models and is easy to study.

2) BERT - BERT is a language model designed to understand text rather than generate long passages. It is strong at tasks such as classification, question answering, and extracting meaning from sentences. On Hugging Face, BERT models are popular because they can be fine-tuned for many practical text tasks.

3) Stable Diffusion - Stable Diffusion is an open image-generation model family available on Hugging Face. It creates images from text prompts, so it belongs to generative AI. People use it for design ideas, artistic images, and visual experimentation because it can produce detailed pictures from short descriptions.

Q4. Difference between traditional programming and AI-assisted development

Traditional programming works by writing exact rules. The developer tells the computer what inputs to look for and what output to produce. If the rules are wrong or incomplete, the program behaves incorrectly. In AI-assisted development, developers still guide the work, but AI tools help by suggesting code, explaining errors, writing test cases, summarizing files, and speeding up routine work. Traditional programming depends on fixed logic written by humans, while AI-assisted development combines human direction with model-generated help.

Q5. Explain AI hallucinations with one example

An AI hallucination happens when a model gives an answer that sounds confident but is not true. This can happen because the model is predicting likely text, not checking facts like a database. For example, a chatbot might invent a research paper title, a fake author name, or a non-existent website and present it as real. That is why important answers from AI should be checked against trusted sources.

Prepared in original wording for academic use.